Class-Domain Incremental Segmentation for Remote Sensing Images

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***Abstract*—Significant progress has been made in class-incremental (learning new classes without forgetting old ones) and domain-incremental (adapting to data from different distributions) semantic segmentation for remote sensing (RS) images. However, in real-world deployment, class-space changes and distribution shifts may co-occur between old and new data. Existing incremental learning methods typically address only one type of shift, struggling to handle joint class and domain incremental learning. To achieve class-domain incremental segmentation, we propose CDISeg, a novel framework that enables cross-domain knowledge accumulation through feature synthesis. CDISeg employs a temporary style encoder while repurposing the segmentation model's backbone as the content encoder. By enforcing orthogonality between their outputs, the model disentangles image content from style, thereby preserving domain-specific style features throughout incremental learning steps. The framework synthesizes old-domain features by projecting old-domain styles onto new-domain content, which supports the preservation of old knowledge, extends new classes to past domains, and facilitates the learning of old classes over new-domain images. Additionally, we introduce class-aware style randomization (CaSR) to enhance feature disentanglement and improve synthesis quality. Extensive experiments on ISPRS and OpenEarthMap datasets demonstrate the remarkable superiority of CDISeg in enabling models to progressively acquire new classes from new domains while recognizing all learned classes across all encountered domains.**

***Index Terms*—Class-incremental learning, domain-incremental learning, feature synthesis, remote sensing (RS) image, semantic segmentation.**

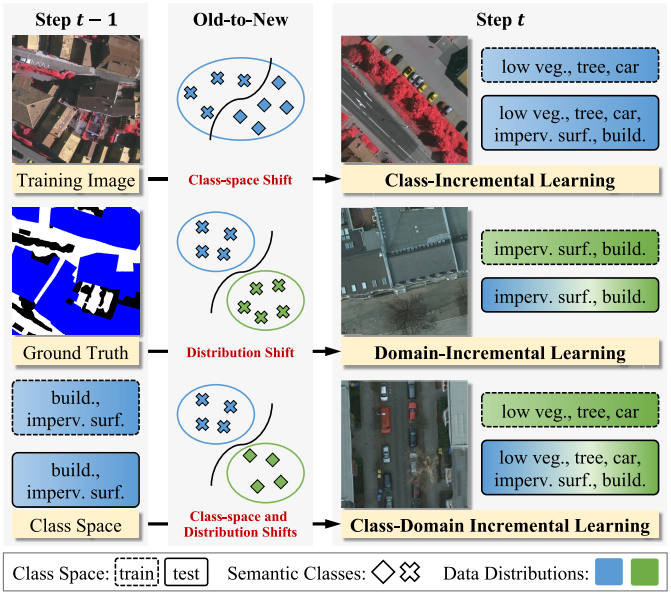


Fig. 1. Illustration about different incremental settings for semantic segmentation of RS images. Class-incremental learning enables models to learn new classes while retaining performance on previous ones. Domain-incremental learning adapts to changing data distributions without altering the class set. Class-domain incremental learning unifies both challenges, allowing models to progressively acquire new classes from new domains while recognizing all learned classes across all encountered domains, a setting reflective of real-world deployment.

I. Introduction

INCREMENTAL segmentation for remote sensing (RS) images aims to develop segmentation models capable of continuously acquiring knowledge from sustained Earth observation data without forgetting previous knowledge, ensuring adaptability in open-world environments with dynamically evolving requirements [1], [2], [3]. Based on the incremental setting, existing research can be broadly categorized into class-incremental and domain-incremental semantic segmentation. As illustrated in Fig. 1, class-incremental semantic segmentation enables models to progressively learn new semantic categories while distinguishing all learned classes during inference [4], [5], [6]. In contrast, domain-incremental semantic segmentation focuses on learning a fixed set of classes across different data distributions [7], [8], [9].

While significant progress has been achieved in class-incremental [10], [11], [12], [13] and domain-incremental [8], [14], [15] semantic segmentation for RS images, their incremental settings deviate from practical deployment scenarios, where distribution shift and class-space shift may co-occur between old and new data. For example, RS images acquired in Africa demonstrate both significant domain gaps compared to those from Europe and the need to recognize novel semantic categories (e.g., large-scale deserts). Furthermore, sensors with distinct spectral band configurations serve different application purposes, leading to their data exhibiting both distribution and class-space variations. Consequently, there is an imperative to develop incremental learning methods that enable models to progressively learn new classes from new domains while recognizing all learned classes across all encountered domains during inference, a novel incremental setting we term *class-domain incremental learning*.

In this article, we address class-domain incremental segmentation for RS images, as exemplified by the bottom right panel of Fig. 1. When only new-class data from a new

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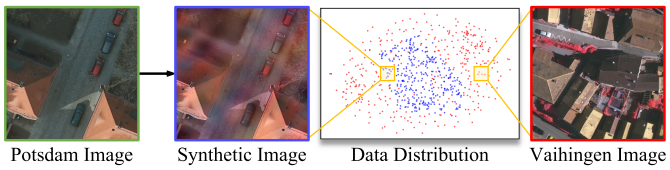


Fig. 2. T-SNE [16] visualization of the data distribution for synthetic and real images generated by LwS [17] under the incremental setup, where three classes are first learned on Vaihingen, followed by two classes on Potsdam. LwS uses the stored Vaihingen domain statistics and Potsdam images to synthesize Vaihingen images. A clear distribution gap can be observed between the synthetic and real images, which hinders the model's ability to maintain old knowledge learned on Vaihingen and also impedes the transfer of new classes to the Vaihingen domain.

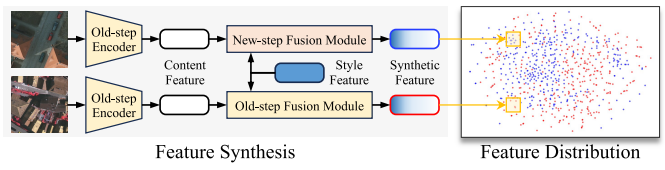


Fig. 3. T-SNE visualization of distribution comparison for synthetic and real Vaihingen features generated by CDISeg under the incremental setup, where three classes are first learned on Vaihingen, followed by two classes on Potsdam. CDISeg synthesizes Vaihingen features by projecting the style features from Vaihingen onto the content features from Potsdam. The mixture of synthetic features with real features (composed of Vaihingen's content and style) indicates their close proximity, facilitating the model's cross-domain knowledge accumulation.

domain becomes available, the segmentation model faces the simultaneous requirements of learning new classes from the new domain while preserving old-class knowledge in old domains, adapting old-class knowledge to the new domain, and extending new-class knowledge to past domains. To achieve this, pioneering LwS work [17] employs style transfer [18] and knowledge distillation [19] to derive effective solutions. The framework first synthesizes images by transferring statistical properties from all encountered domains to the available images. Then, the new model is regularized to mimic the old model's predictions on these synthesized images, preserving knowledge of old classes from past domains and progressively accumulating it in the new one. Concurrently, it is trained on the synthesized images paired with new-class annotations to acquire cross-domain knowledge for new classes.

LwS, while effective in class-domain incremental segmentation for natural images, proves inadequate for RS images. This is due to an inevitable domain gap between synthesized images (generated from stored domain-specific statistics) and real RS images, a consequence of the latter's inherent complexity, as evidenced in Fig. 2. This inevitable domain gap negatively impacts the new model's learning in two key aspects. First, it renders the old model incapable of producing reliable predictions on synthesized images, which compromises the effectiveness of knowledge distillation and leads to the new model's catastrophic forgetting of old classes. Second, it prevents the new model from effectively extending new-class knowledge to past domains. Although advanced generative models (e.g., RS-SD [20] and CRS-Diff [21]) can improve the realism of synthesized images, employing a dedicated generator for each new domain remains impractical. An alternative approach using a single generative model would, nonetheless, require its own incremental learning capability to adapt to evolving data distributions. This introduces a parallel challenge of developing effective incremental learning methods for generative models [22], [23].

Unlike LwS, which resorts to explicit image synthesis, we introduce feature synthesis to circumvent the negative impacts of domain gaps between synthesized and real images. Specifically, at each learning step, we disentangle the feature representation of available images into content and style features, maintaining domain-specific style characteristics. By projecting old-domain style features onto content features of new-domain images, we synthesize old-domain features. A comparison of the distributions between synthesized (blue dots) and real (red dots) old-domain features, presented in Fig. 3, reveals no clear separation between the two groups. This indicates that the synthesized and real features lie in proximity within the feature space. Such alignment enables the old model to correctly recognize old classes in new-domain images, thereby benefiting both preservation and adaptation of old-class knowledge. Moreover, these aligned synthesized features facilitate the extension of new-class knowledge to past domains.

To materialize this idea, we propose CDISeg, a novel learning framework that effectively captures domain-specific style features and incrementally accumulates class knowledge across domains. The framework employs a temporary style encoder alongside the segmentation model, repurposing the model's backbone as the content encoder. Through orthogonal constraints on the feature representations of both encoders, CDISeg achieves image content-style disentanglement. Then, the domain-specific style is computed by averaging style features extracted from available training images and stored in a memory bank. During subsequent learning steps, these stored styles are fused with new-domain content to reconstruct old-domain representations for replay. Building upon this architecture, CDISeg, following LwS, employs four specialized loss terms to learn new classes across the new and past domains, retain old-class knowledge over past domains, and adapt old-class knowledge to the new domain, respectively.

The devised framework's performance critically depends on the quality of content-style disentanglement, which directly affects old-domain feature synthesis. However, simple orthogonality constraints cannot ensure meaningful separation where content features represent high-level semantics while style features capture low-level appearance (e.g., texture and color). To address this limitation, we present class-aware style randomization (CaSR), generating randomly stylized versions of training images while enforcing content feature consistency between stylized and original images. This strategy guides the content encoder to concentrate on semantic extraction rather than style variations. Importantly, our stylized images differ from those in LwS: ours inject random styles into new-domain images to reinforce content-style disentanglement, while LwS transfers preserved old-domain statistics to new-domain images exclusively for memory replay.

To evaluate the effectiveness of CDISeg, we employ the ISPRS[1] and OpenEarthMap [24] datasets, selected for their pronounced distribution shifts stemming from distinct sensor

[1]https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/

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band configurations (ISPRS) and geographic imaging locations (OpenEarthMap). These inherent domain gaps facilitate the design of novel experimental setups that simulate realistic incremental learning scenarios where both distribution shift and class-space shift exist between old and new RS data. Compared to state-of-the-art methods, such as LwS and MiSSNet [13], CDISeg consistently demonstrates superior performance across all experimental setups. Under three different setups on the ISPRS dataset, CDISeg achieves performance gains of **1.89%**, **7.78%**, and **4.90%**, respectively.

The main contributions of this article are summarized as follows.

1) We propose a novel learning framework for class-domain incremental segmentation of RS images, dubbed CDISeg, which resorts to image content-style disentanglement coupled with old-domain feature synthesis, achieving accumulation of class knowledge across domains.
2) We introduce CaSR to enhance content-style disentanglement. This module generates randomly stylized versions of training images while constraining content feature consistency between stylized and original images, thereby directing the content encoder to focus on semantic extraction.
3) We devise novel experimental setups that simulate joint class and domain incremental scenarios, while providing strong performance baselines for these challenging setups.

The rest of the article is organized as follows. Section II reviews the related works. Section III elaborates on the details of the CDISeg framework. Section IV reports experimental results and provides analysis. Finally, conclusions are drawn in Section V.

II. Related Work

Current research in RS addresses class-incremental and domain-incremental semantic segmentation as separate challenges, while their joint learning scenario remains unexplored. Accordingly, this section reviews key advancements in each incremental setting individually.

A. Class-Incremental Segmentation in RS

Class-incremental segmentation is the predominant focus in RS incremental segmentation research [1]. The long-distance observation nature of RS sensors causes newly acquired images to frequently contain old-class objects (labeled as background in the current learning step). This triggers catastrophic forgetting, where segmentation models misclassify old-class objects as background when learning new classes. Existing solutions primarily leverage knowledge distillation, which forces the new model to retain old-class knowledge by mimicking the old model's predictions on new images, such as its multiscale features [25], class probability distributions [25], [26], or cross-image semantic relationships [12].

Considering characteristics of RS images or segmentation tasks, specialized distillation techniques, such as contrastive distillation [26], diversity distillation [27], task-wise distillation [28], and local semantic distillation [13], have been developed to improve retention of old knowledge further. Among these, contrastive distillation [26] is inspired by rotation invariance in RS images, minimizing the discrepancy between the new model's features (extracted from transformed images) and the old model's transformed features (derived from nonaugmented images). Meanwhile, local semantic distillation [13] addresses the semantic shift of the background class, a challenge unique to semantic segmentation, by aligning the correlation patterns between image features and old-class prototypes across the new and old models. Unlike most works that focus predominantly on knowledge retention, the most recent work MiR [6] addresses three key challenges in class-incremental semantic segmentation: representation bias, interstep imbalance, and interstep similarity, thereby enhancing the new model's ability to learning new classes without forgetting old knowledge.

Despite these advances, existing works only address class-space shifts across different learning steps, restricting them to domain-specific category learning. Consequently, they fail to generalize for cross-domain recognition and remain inadequate for real-world applications in open environments.

B. Domain-Incremental Segmentation in RS

Very few works address domain-incremental segmentation for RS images. The earliest attempt [14] employs gradient boosting for domain-incremental learning, training a base learner for each new domain while freezing previous learners to preserve prior knowledge. Critically, the newly introduced learner serves to compensate for performance limitations of existing learners on new domains, as opposed to operating as a standalone segmentation model dedicated to specific domains (i.e., the conventional nonincremental method requiring separate models per domain). Such a solution is later adopted in [8] for postdisaster building damage assessment. More recently, the GRSM-RS-DIL [15] model addresses catastrophic forgetting without expanding network parameters, instead employing multilevel knowledge distillation and frozen shallow encoder layers.

While these works demonstrate initial progress, their assumption of a static class space throughout incremental learning steps leaves the critical challenge of cross-domain novel class acquisition unresolved. To bridge this gap, we propose feature-level synthesis of old-domain information to enable comprehensive knowledge accumulation across both learned and emerging categories in evolving domains. The resulting segmentation model achieves recognition of all learned semantic categories across all encountered domains.

III. Proposed Method

A. Problem Definition

Class-domain incremental semantic segmentation involves training the model over multiple learning steps $t \in \{1, 2, \ldots, T\}$, enabling it to recognize a progressively expanding set of semantic classes and handle increasingly diverse image domains. At learning step t, the model learns from training dataset $\mathcal{D}^t = \{(\mathbf{X}^t, \mathbf{Y}^t)\} \subset \mathcal{X}^t \times \mathcal{Y}^t$, where $\mathbf{X}^t \in \mathcal{X}^t$ represents an RS image from a new image domain $\mathcal{X}^t$ and $\mathbf{Y}^t \in \mathcal{Y}^t$ is its corresponding ground-truth map. Here, $\mathcal{C}^t$

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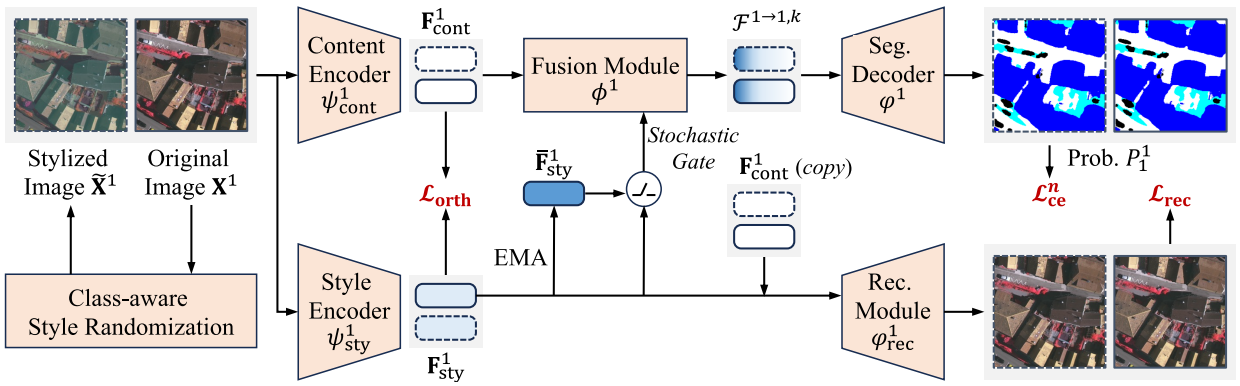


Fig. 4. Overall framework of our proposed CDISeg at the first learning step. CDISeg employs a temporary style encoder while repurposing the model backbone as the content encoder. Orthogonal constraints are applied to enable image content-style disentanglement. Domain-specific styles $\bar{\mathbf{F}}^1_{\text{sty}}$, averaged from the style features $\mathbf{F}^1_{\text{sty}}$ of training images, are fused with new-domain content in later steps to reconstruct old-domain representations, enabling cross-domain knowledge accumulation. CaSR enhances disentanglement by generating stylized images while preserving content consistency. $\mathcal{L}_{\text{orth}}$, $\mathcal{L}^n_{ce}$, and $\mathcal{L}_{\text{rec}}$, respectively, denote the losses for content-style orthogonality, new-domain new-class learning, and stylized-original content consistency. $\mathcal{F}^{1\to1,k}$ represents the synthesized features at the kth training iteration obtained by projecting $\mathbf{F}^1_{\text{sty}}$ or $\bar{\mathbf{F}}^1_{\text{sty}}$ onto the content features $\mathbf{F}^1_{\text{cont}}$.

denotes the set of semantic classes introduced at step t, and each pixel in $\mathbf{Y}^t$ is labeled with a class $c \in \mathcal{C}^t$. Crucially, any objects belonging to previous classes $\mathcal{C}^{1:t-1}$ or future classes $\mathcal{C}^{t+1:T}$ that appear in $\mathbf{X}^t$ are labeled as background (c_b) in $\mathbf{Y}^t$, ensuring disjoint class sets across steps (i.e., $\mathcal{C}^1 \cap \mathcal{C}^2 \cap \cdots \cap \mathcal{C}^T = \emptyset$). By the completion of the step t, the model Φ^t acquires the ability to segment all seen classes $\mathcal{C}^{1:t}$ on all encountered domains $\mathcal{X}^{1:t}$.

B. Feature Disentanglement and Synthesis

Since the segmentation model has access only to current-domain RS images and their corresponding new-class annotations at each step, it faces three challenges during joint class and domain incremental learning as follows.

1) Learning new classes across past domains and the current domain, despite the unavailability of old-domain images annotated with new classes.
2) Preserving segmentation capability for previously learned classes on past domains, without access to their original annotated training data.
3) Adapting old-class knowledge to the current domain, even when these classes are annotated as background in the current training data.

To tackle these challenges, we draw inspiration from LwS [17] to synthesize old-domain information, thereby enabling cross-domain knowledge accumulation. Accordingly, we propose CDISeg, a novel class-domain incremental learning framework specifically designed for semantic segmentation of RS images, depicted in Fig. 4. Rather than explicitly synthesizing old-domain images, we adopt a feature synthesis strategy for replaying old-domain information. This strategy preserves domain-specific style features by image content-style disentanglement and then projects them onto the content features of the current domain to produce old-domain representations. To implement this strategy, CDISeg incorporates a temporary style encoder, discarded after each learning step to avoid additional inference latency in the final deployed model, while repurposing the segmentation backbone as the content encoder. This architecture naturally suggests an intuitive content-style disentanglement method by imposing orthogonality constraints between both encoders' feature representations [29].

1) Feature Disentanglement: Formally, at step t, let ψ^t_{cont} and ψ^t_{sty} denote the content and style encoders, respectively. Given an accessible RS image $\mathbf{X}^t \in \mathcal{D}^t$, the content and style features are extracted by $\mathbf{F}^t_{\text{cont}} = \psi^t_{\text{cont}}(\mathbf{X}^t) \in \mathbb{R}^{h\times w\times d}$ and $\mathbf{F}^t_{\text{sty}} = \psi^t_{\text{sty}}(\mathbf{X}^t) \in \mathbb{R}^{1\times 1\times d}$, where $h \times w$ represents the spatial dimensions and d the channel dimension. The orthogonality constraint is enforced by minimizing the following loss:

$$\mathcal{L}_{\text{orth}} = \frac{1}{|\mathcal{D}^t|} \sum_{\mathbf{X}^t \in \mathcal{D}^t} \left| \text{CoS}(\text{GAP}(\mathbf{F}^t_{\text{cont}}), \mathbf{F}^t_{\text{sty}}) \right| \tag{1}$$

where $\text{CoS}(\mathbf{u}, \mathbf{v}) = (\langle \mathbf{u}, \mathbf{v} \rangle)/(\|\mathbf{u}\| \cdot \|\mathbf{v}\|)$ is cosine similarity and $\text{GAP}(\mathbf{F}) = 1/(hw) \sum_{i=1}^{h} \sum_{j=1}^{w} \mathbf{F}_{i,j,:}$ denotes global average pooling. Since the style encoder extracts spatially agnostic style characteristics, we design it to output feature maps with a spatial dimension of 1×1. In contrast, the content feature maintains higher spatial resolution for subsequent segmentation. Enforcing rigorous pixel-wise orthogonality constraints between content and style features not only introduces optimization challenges but also potentially compromises the model's ability to learn discriminative features for accurate segmentation. We therefore implement global average pooling on content features and impose orthogonality at the global representation level between style and content.

2) Domain-Specific Style Features: Having achieved feature disentanglement, we derive domain-specific style characteristics $\bar{\mathbf{F}}^t_{\text{sty}}$ via exponential moving average (EMA) with momentum $\beta = 0.999$, updated iteratively during training as

$$\bar{\mathbf{F}}^{t,k}_{\text{sty}} = \beta \bar{\mathbf{F}}^{t,k-1}_{\text{sty}} + (1-\beta)\mathbf{F}^{t,k}_{\text{sty}} \tag{2}$$

where $\mathbf{F}^{t,k}_{\text{sty}}$ is the style feature extracted from input $\mathbf{X}^t$ at the kth training iteration. The resultant domain-specific style feature for incremental step t is obtained as $\bar{\mathbf{F}}^t_{\text{sty}} = \bar{\mathbf{F}}^{t,K}_{\text{sty}}$, where K is the total iterations at step t, and stored in a progressively filled memory bank $\mathcal{M}^{1:t-1} = \{\bar{\mathbf{F}}^m_{\text{sty}} | m < t\}$. This update strategy avoids costly full-dataset recomputation while enhancing segmentation accuracy, as will be analyzed.

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3) Semantic Segmentation: In principle, content features capture high-level semantics while style features encode low-level visual appearance (e.g., color). Although content features alone could be sufficient for segmentation when fed directly into a decoder φ^t, RS images are unique: color variations in ground objects may originate from their intrinsic spectral characteristics, indicating that colors correlate with land-cover categories and can serve as a discriminative feature for object identification. Motivated by this fact, we propose fusing content and style features before decoder processing, leveraging comprehensive image information for precise segmentation. During inference, the domain-specific style is projected onto the input's content features, as the style encoder is discarded posttraining. In contrast, the training phase employs a stochastic fusion strategy that selects either the domain-specific style or the current batch's style features with equal probability (50%). This design stems from empirical observations that such random style selection enhances the decoder's generalization capability, leading to improved segmentation performance. Formally, the decoder's input feature $\mathcal{F}^{t,k}$ during training is computed as

$$\mathcal{F}^{t,k} = \phi^t(\mathbf{F}^{t,k}_{\text{cont}}, \hat{\mathbf{F}}^{t,k}_{\text{sty}}) = \text{ReLU}(\text{AdaIN}(\text{Conv}(\mathbf{F}^{t,k}_{\text{cont}}), \hat{\mathbf{F}}^{t,k}_{\text{sty}})) \quad (3)$$

where

$$\hat{\mathbf{F}}^{t,k}_{\text{sty}} = \begin{cases} \bar{\mathbf{F}}^{t,k-1}_{\text{sty}}, & \mathbb{P} = 0.5 \\ \mathbf{F}^{t,k}_{\text{sty}}, & \mathbb{P} = 0.5. \end{cases} \quad (4)$$

Here, ϕ^t denotes the fusion module composed of convolution, AdaIN [30], and ReLU operations in sequence.

4) Feature Synthesis: RS images, captured from a distance, often contain objects spanning past, present, and future classes. However, due to variations in imaging conditions (e.g., sensor parameters and geographic locations), the visual appearance of these objects may differ. Based on this assumption, we leverage the memory bank $\mathcal{M}^{1:t-1}$ and fusion module ϕ^t to synthesize old-domain features by transferring style features from past domains onto content features of RS images accessible at step t, as

$$\mathcal{F}^{t\to m,k} = \phi^t(\mathbf{F}^{t-1,k}_{\text{cont}}, \bar{\mathbf{F}}^{m}_{\text{sty}}) \quad (5)$$

where $\mathcal{F}^{t\to m,k}$ represents old-domain features at step $m (m < t)$, with the superscript $t \to m$ indicating feature synthesis from step t to m. Crucially, since CDISeg synthesizes features (not images), it cannot preprepare old-domain data before segmentation model training, unlike LwS. Instead, CDISeg conducts old-domain feature synthesis in each training iteration (denoted by superscript k) when learning new classes. For consistency, we update $\mathcal{F}^{t,k}$ in (3) to $\mathcal{F}^{t\to t,k}$. Both $\mathcal{F}^{t\to t,k}$ and $\mathcal{F}^{t\to m,k}$ are synthesized using semantic information from the same image. However, the former employs content features $\mathbf{F}^{t,k}_{\text{cont}}$ extracted from $\mathbf{X}^t$ by ψ^t_{cont}, while the latter utilizes $\mathbf{F}^{t-1,k}_{\text{cont}}$ extracted by ψ^{t-1}_{cont}. This design is intentional: content and style features are constrained to be orthogonal at each learning step, while content features from the current model (step t) may not be orthogonal to style characteristics of previous domains. Using the content encoder from step $t-1$ helps ensure that the synthesized old-domain features align more closely with the real feature distribution of past domains. With both new-domain and old-domain features now available, the framework is prepared for cross-domain knowledge accumulation.

C. Class-Domain Incremental Learning

The old-domain features, which reiterate old-domain distributions, enable learning of new classes across past domains while preserving knowledge of old classes. Concurrently, the new-domain features support learning of new classes from the current domain and adaptation of old classes to the new domain. The ground truth $\mathbf{Y}^t$ can supervise the segmentation model to learn new classes over both new and past domains, but fails to provide supervision for old-class preservation on old-domain features or old-class adaptation on new-domain features, since old-class objects are labeled as background in $\mathbf{Y}^t$. To address this, we employ the segmentation model from step $t-1$ to produce pseudo-labels for old classes.

At the kth training iteration of step t, while the new-domain RS image $\mathbf{X}^t$ is processed by the current segmentation model, it is simultaneously fed to the old-step one exclusively for old-class prediction as follows:

$$P^{t-1}_m = \varphi^{t-1}(\phi^{t-1}(\psi^{t-1}_{\text{cont}}(\mathbf{X}^t), \bar{\mathbf{F}}^m_{\text{sty}})) \quad (6)$$

where $P^{t-1}_m \in \mathbb{R}^{H\times W\times(|\mathcal{C}^{1:t-1}|+1)}$ represents the classification probability predicted by the old-step segmentation model for the new-domain image under the style of step m. ψ^{t-1}_{cont}, φ^{t-1}, and ϕ^{t-1} denote the encoder, decoder, and fusion module of the old-step model, respectively. By step t, the memory bank stores $t-1$ previously learned styles, and the old-step model produces $t-1$ such predictions P^{t-1}_m (one per style), which are collectively denoted as $P^{t-1}_{<t}$ for all old classes $\mathcal{C}^{1:t-1}$. With the new-class ground truth $\mathbf{Y}^t$ and old-class predictions $P^{t-1}_{<t}$ established, CDISeg employs four specialized loss terms to achieve class knowledge accumulation across new and past domains, as shown in Fig. 5.

1) Learning New Classes From the New Domain: By inputting the new-domain feature $\mathcal{F}^{t\to t,k}$ into the decoder φ^t, we can obtain the current segmentation model's classification probability $P^t_t \in \mathbb{R}^{H\times W\times|\mathcal{C}^{1:t}|+1}$, where the superscript t indicates the prediction generated by the model at step t, while the subscript t denotes that this prediction correspond to the new-domain image under the style of step t. To address the background shift peculiar to class-incremental semantic segmentation, we adopt the background-modeled cross-entropy loss proposed in MiB [31], [32] for effective learning of new classes from the new domain. The loss function is formulated as

$$\mathcal{L}^n_{ce} = -\frac{1}{|\mathcal{D}^t|}\sum_{\mathbf{X}^t,\mathbf{Y}^t\in\mathcal{D}^t} \mathbf{Y}^t \cdot \log \mathring{P}^t_t \quad (7)$$

where

$$\mathring{P}^t_t[x,y,c] = \begin{cases} P^t_t[x,y,c], & \text{if } \mathbf{Y}^t[x,y] \neq c_b \\ \sum_{c'\in\acute{\mathcal{C}}^{1:t-1}} P^t_t[x,y,c'], & \text{if } \mathbf{Y}^t[x,y] = c_b. \end{cases} \quad (8)$$

$\acute{\mathcal{C}}^{1:t-1} = \mathcal{C}^{1:t-1} \cup \{c_b\}$ represents the extended class space encompassing all old classes and the background class, while $P^t_t[x,y,c]$ represents the model's predicted probability for class c at pixel location (x,y) in the image $\mathbf{X}^t$. The superscript n in $\mathcal{L}^n_{ce}$ emphasizes that these predictions are generated under the new style feature.

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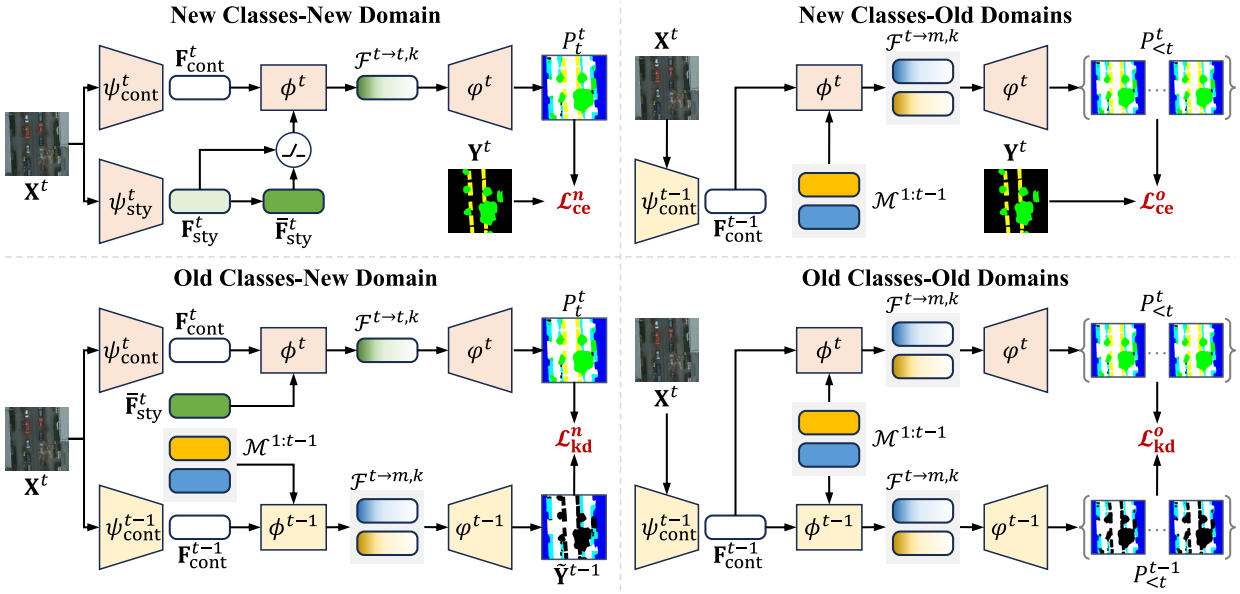


Fig. 5. Cross-domain knowledge accumulation at incremental step ($t > 1$). As in the step 1 illustration in Fig. 4, CDISeg incorporates a temporary reconstruction module and CaSR, omitted here for clarity. Using new-domain images $\mathbf{X}^t$ and old-domain style memory bank $\mathcal{M}^{1:t-1}$, it synthesizes old- and new-domain features ($\mathcal{F}^{t\to m,k}$ and $\mathcal{F}^{t\to t,k}$), where the superscript denotes feature synthesis from step t to m at the kth training iteration. Then, four losses guide model learning: $\mathcal{L}^n_{ce}$ (acquire new classes), $\mathcal{L}^o_{ce}$ (extend new classes to old domains), $\mathcal{L}^n_{kd}$ (adapt old classes to the new domain), and $\mathcal{L}^o_{kd}$ (preserve old classes over past domains). $P^t_{<t}$ and $P^{t-1}_{<t}$ denote the predictions of the step-t and step-$(t-1)$ models, respectively, on the new-domain image rendered with the $t-1$ old-domain styles.

2) Extending New Classes to Past Domains: As described earlier, feature synthesis generates $t-1$ old-domain features $\mathcal{F}^{t\to m,k}$ for the input $\mathbf{X}^t$ at the kth iteration. These features are individually processed by the decoder φ^t for segmentation. The model learns new classes over past domains by applying supervision using new-class ground truth maps on predictions derived from these old-domain features. This is achieved by optimizing

$$\mathcal{L}^o_{ce} = -\frac{1}{|\mathcal{D}^t|}\sum_{\mathbf{X}^t,\mathbf{Y}^t\in\mathcal{D}^t}\frac{1}{t-1}\sum_{m=1}^{t-1}\mathbf{Y}^t\cdot\log\mathring{P}^t_m \tag{9}$$

where the modified prediction $\mathring{P}^t_m$ is computed from P^t_m via (8) to handle the background shift. The superscript o in $\mathcal{L}^o_{ce}$ indicates that these predictions are derived from synthesized features with old styles.

3) Adapting Old Classes to the New Domain: While the old-step model offers supervision to guide the current model in adapting old classes to the new domain, deriving reliable pseudo-labels from old-model predictions $P^{t-1}_{<t}$ (one per past style) is nontrivial. We adopt the hard pseudo-labeling strategy from [17], formulated as

$$\ddot{\mathbf{Y}}^{t-1}[x,y] = \arg\max_{c\in\acute{\mathcal{C}}^{1:t-1}}\max_{m<t}P^{t-1}_m[x,y] \tag{10}$$

where $\max_{m<t}P^{t-1}_m[x,y]$ selects the probability vector with maximum peak value for each pixel (x,y) across old styles. To mitigate noise in pseudo-labels, we refine them using the new-class ground-truth map $\mathbf{Y}^t$

$$\tilde{\mathbf{Y}}^{t-1} = \begin{cases}\ddot{\mathbf{Y}}^{t-1}, & \text{if } \Gamma(\ddot{\mathbf{Y}}^{t-1})\wedge\mathbf{Y}^t = c_b\\ c_b, & \text{if } \mathbf{Y}^t\neq c_b\\ \text{ignore}, & \text{otherwise}\end{cases} \tag{11}$$

where $\tilde{\mathbf{Y}}^{t-1}$, $\ddot{\mathbf{Y}}^{t-1}$, and $\mathbf{Y}^t$ omit $[x,y]$ for notational simplicity. $\Gamma(\cdot)$ filters $\ddot{\mathbf{Y}}^{t-1}$ to retain only confident predictions, with two criteria: the peak probability $P^{t-1}_{\max}\left[x,y,\ddot{\mathbf{Y}}^{t-1}[x,y]\right]$ exceeds a threshold τ, or it ranks among the top-N fraction of highest peaks for class $\ddot{\mathbf{Y}}^{t-1}[x,y]$. Here, $P^{t-1}_{\max} = \max_{m<t}P^{t-1}_m$, with $\tau = 0.9$ and $N = 0.66$ as in [17].

To train the model using these refined pseudo-labels, we minimize the following objective:

$$\mathcal{L}^n_{kd} = -\frac{1}{|\mathcal{D}^t|}\sum_{\mathbf{X}^t\in\mathcal{D}^t}\tilde{\mathbf{Y}}^{t-1}\cdot\log\dot{P}^t_t \tag{12}$$

where $\dot{P}^t_t$ is the adjusted classification probability of the current model, accounting for the background shift

$$\dot{P}^t_t[x,y,c] = \begin{cases}P^t_t[x,y,c], & \text{if } \tilde{\mathbf{Y}}^{t-1}[x,y]\neq c_b\\ \sum_{c'\in\acute{\mathcal{C}}^t}P^t_t[x,y,c'], & \text{if } \tilde{\mathbf{Y}}^{t-1}[x,y]=c_b\end{cases} \tag{13}$$

and $\acute{\mathcal{C}}^t = \mathcal{C}^t\cup\{c_b\}$.

4) Preserving Old Classes Over Past Domains: The model's absorption of new-domain or new-class knowledge rewrites weights, leading to the forgetting of old classes over past domains. To mitigate this, we employ knowledge distillation that enforces the current model to mimic the old-step model's predictions on old classes. While this process shares similarities with guiding the current model to adapt old classes to the new domain, a key distinction lies in our use of soft pseudo-labels instead of hard labels. The motivation behind this design is that the distribution of synthesized old-domain features cannot achieve perfect alignment with real old-domain features, inevitably resulting in classification errors in old-model predictions for new-domain images under old styles. Fundamentally, old-class adaptation involves learning new knowledge. To ensure the newly learned knowledge is as

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accurate as possible, we employ an elaborate pseudo-label generation strategy. In contrast, the objective of old-class preservation is to prevent modification of existing knowledge. Therefore, we directly use the old-class predictions $P_{<t}^{t-1}$ to supervise the current model's training, formulated as

$$\mathcal{L}_{kd}^{o} = -\frac{1}{|\mathcal{D}^t|}\sum_{\mathbf{X}^t\in\mathcal{D}^t}\frac{1}{t-1}\sum_{m=1}^{t-1}P_m^{t-1}\cdot\log\dot{P}_m^t \tag{14}$$

where $\dot{P}_m^t$ is computed from P_m^t using a derivation process analogous to (13)

$$\dot{P}_m^t[x,y,c] = \begin{cases} P_m^t[x,y,c], & \text{if } c\in\mathcal{C}^{1:t-1} \\ \sum_{c'\in\acute{\mathcal{C}}^t} P_m^t[x,y,c'], & \text{if } c=c_b. \end{cases} \tag{15}$$

The training objective for the current model to accomplish cross-domain knowledge accumulation is formulated as

$$\mathcal{L}_{\text{cka}} = \mathcal{L}_{ce}^{n} + \lambda_{ce}^{o}\cdot\mathcal{L}_{ce}^{o} + \lambda_{kd}^{n}\cdot\mathcal{L}_{kd}^{n} + \lambda_{kd}^{o}\cdot\mathcal{L}_{kd}^{o} \tag{16}$$

where λ_{ce}^{o}, λ_{kd}^{n}, and λ_{kd}^{o} are tunable hyperparameters to balance the loss terms. Following [17], we set $\lambda_{ce}^{o} = \lambda_{kd}^{n} = \lambda_{kd}^{o} = 10$, and keep them fixed across all experiments. Although this choice is empirical, it is well-motivated: as incremental learning progresses, the model encounters a growing number of classes and domains. Accordingly, we assign higher weights to losses associated with old classes and domains to limit forgetting, while allowing a small performance tradeoff on new-domain classes to maximize overall learning. It should be noted that at the first step ($t = 1$), the absence of old-domain style features causes the training objective to degenerate to a standard cross-entropy loss.

D. Class-Aware Style Randomization

Relying exclusively on orthogonality constraints proves insufficient for achieving effective feature disentanglement, where content features should capture image semantics while style features preserve low-level visual appearance. The degree of feature disentanglement fundamentally governs both the quality of synthesized old-domain features and the ultimate efficacy of our framework. To enhance feature disentanglement, we develop a CaSR module tailored for RS image characteristics. The CaSR module generates stylized variants of new-domain images by randomizing their styles while requiring the content encoder to produce consistent content features between these stylized images and their original counterparts.

Specifically, imaging conditions such as sensor band configuration and geographic location often induce intracategory visual homogeneity in RS images. For instance, objects of the same category share similar spectral characteristics, leading to relatively consistent colors under specific band configurations. In addition, within a geographic region, intraclass objects (e.g., buildings) often display uniform visual patterns due to anthropogenic design. Inspired by this observation, we propose to apply identical style randomization to objects belonging to the same category within an input image, then combine them to generate the final stylized output. For style randomization, we employ convolutional layers with randomly sampled weights to diversify textures, followed by replacing the original style statistics with randomized ones to achieve stylistic variation [33], [34].

1) Stylized Image Generation: Formally, intraclass objects within a new-domain image are calculated as $\mathbf{X}_c^t = \mathbf{X}^t\cdot\mathbf{M}_c$, where $\mathbf{M}_c = \mathbb{I}(\mathbf{Y}^t = c)$ is the binary mask for class c. The style randomization for $\mathbf{X}_c^t$ consists of two sequential processes. First, a recursive texture modification is applied through l random convolutional layers

$$\ddot{\mathbf{X}}_c^t = \text{RC}(\mathbf{X}_c^t, l) = \begin{cases} \text{RC}(\mathbf{X}_c^t, l-1)*\Theta_l, & \text{if } l>0 \\ \mathbf{X}_c^t, & \text{if } l=0 \end{cases} \tag{17}$$

where l is sampled from $\{1,2,\ldots,l_{\max}\}$, where we follow [34] in setting $l_{\max} = 10$ to achieve sufficient texture variation without compromising semantics. Each convolutional layer's weights $\Theta_l \in \mathbb{R}^{s_l\times s_l\times 3\times 3}$ are drawn from a Gaussian distribution $\mathcal{N}(0, 1/3s_l^2)$, with kernel size s_l randomly selected from $\{1,3,5,7\}$. $*$ is the convolution operation. Subsequently, the style transformation is performed by

$$\tilde{\mathbf{X}}_c^t = \sigma'\cdot\frac{\ddot{\mathbf{X}}_c^t - \mu(\ddot{\mathbf{X}}_c^t)}{\sigma(\ddot{\mathbf{X}}_c^t)} + \mu'. \tag{18}$$

$\sigma(\cdot)$ and $\mu(\cdot)$ compute the channel-wise standard deviation and mean of the input, respectively. The random statistics σ' and μ' are initialized by sampling from a standard normal distribution $\mathcal{N}(0,1)$. To mitigate potential semantic distortion, we perform a weighted combination between the original and stylized versions for each class using randomly sampled mixing ratios $\alpha\in[0,1)$. The final stylized image $\tilde{\mathbf{X}}^t$ is obtained through the aggregation of all classes

$$\tilde{\mathbf{X}}^t = \sum_{c\in\acute{\mathcal{C}}^t}\left(\alpha\mathbf{X}_c^t + (1-\alpha)\tilde{\mathbf{X}}_c^t\right). \tag{19}$$

2) Content Consistency Constraint: The stylized image $\tilde{\mathbf{X}}^t$ and its corresponding original image $\mathbf{X}^t$ are simultaneously processed be the content encoder ψ_{cont}^t to extract their respective content features $\tilde{\mathbf{F}}_{\text{cont}}^t$, $\mathbf{F}_{\text{cont}}^t$. A common way to enforce content consistency is to minimize pixel-wise differences between the two content features. However, the substantial visual discrepancies between stylized and original images make pixel-wise alignment challenging. Moreover, imposing such a strict constraint could impair the encoder's ability to learn high-level semantics. Instead of relying on explicit feature alignment, we propose to establish consistency by requiring that both content features, when combined with the original image's style features, must be capable of reconstructing the original input. This reconstruction-based constraint provides a more flexible consistency measure, maintaining the goal of consistent contents while allowing for natural feature variations induced by style transformation and promoting semantically meaningful feature learning.

To implement the proposed constraint, we introduce a temporary reconstruction module that is jointly trained with the segmentation model and subsequently discarded after training completion. The module receives both content features $\tilde{\mathbf{F}}_{\text{cont}}^t$ and $\mathbf{F}_{\text{cont}}^t$, along with the original style feature $\mathbf{F}_{\text{sty}}^t$ as input, generating reconstructed versions of the original image.

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This leads to the following consistency loss function that we optimize:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{D}^t|}\sum_{\mathbf{X}^t\in\mathcal{D}^t}\left|\varphi^t_{\text{rec}}(\mathbf{F}^t_{\text{cont}},\mathbf{F}^t_{\text{sty}})-\mathbf{X}^t\right| + \frac{1}{|\mathcal{D}^t|}\sum_{\mathbf{X}^t\in\mathcal{D}^t}\left|\varphi^t_{\text{rec}}(\tilde{\mathbf{F}}^t_{\text{cont}},\mathbf{F}^t_{\text{sty}})-\mathbf{X}^t\right| \tag{20}$$

where φ^t_{rec} represents the reconstruction module and $\tilde{\mathbf{F}}^t_{\text{cont}} = \psi^t_{\text{cont}}(\tilde{\mathbf{X}}^t)$. During training, the consistency loss and orthogonality loss in (1) are jointly optimized to achieve effective content-style feature disentanglement. The overall objective $\mathcal{L}_{\text{tot}}$ for the current model is given by

$$\mathcal{L}_{\text{tot}} = \mathcal{L}_{\text{cka}} + \lambda_{\text{orth}}\cdot\mathcal{L}_{\text{orth}} + \lambda_{\text{rec}}\cdot\mathcal{L}_{\text{rec}} \tag{21}$$

where λ_{orth} and λ_{rec} are weighting factors for content-style orthogonality and stylized-original content consistency, respectively. In all experiments, we empirically set $\lambda_{\text{orth}} = 10$ and $\lambda_{\text{rec}} = 0.01$. To maximize the potential of our carefully generated stylized images, we additionally process them through the style encoder to reinforce content-style feature orthogonality. Meanwhile, the stylized content features are fused with both new-domain and old-domain style features for segmentation, facilitating cross-domain knowledge accumulation. Consequently, both $\mathcal{L}_{\text{cka}}$ and $\mathcal{L}_{\text{orth}}$ incorporate computations from original and stylized images, similar to $\mathcal{L}_{\text{rec}}$.

During the inference phase, although we preserve style features from all encountered domains, there is no criterion to determine which domain-specific style features should be projected onto the input's content features. Since our goal is to progressively accumulate semantic category knowledge across all seen domains, we simply use the latest domain's style features for segmentation.

IV. Experiments

A. Experimental Setup

1) Datasets: We conduct experiments on two widely used RS semantic segmentation datasets: ISPRS[1] and OpenEarthMap [24]. These datasets exhibit inherent distributional diversity, which facilitates the simulation of joint incremental learning across classes and domains. To be specific, *ISPRS* consists of two subsets, Vaihingen and Potsdam, each covering a specific geographic region and acquired with distinct sensor band configurations. The Vaihingen subset provides three-channel RS images in near-infrared, red, and green bands, while the Potsdam subset offers red, green, blue, and near-infrared band combinations. Both subsets share an identical label space comprising six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. *OpenEarthMap* consists of aerial and satellite images covering 44 countries across 6 continents. This enables continent-wise data partitioning to evaluate our CDISeg's capability in handling class-domain incremental learning under geographic variations. The dataset focuses on eight semantic classes: bare-land, rangeland, developed space, road, tree, water, agricultural land, and buildings.

TABLE I
Three Experimental Setups for Class-Domain Incremental Semantic Segmentation in RS

Setting	Dataset	Classes $\mathcal{C}^t$	Domains $\mathcal{X}^t$
Changing Band Configuration	Potsdam	{imperv. surf., build.}^{1} {low veg.}^{2} {tree}^{3} {car}^{4}	{IRRG}^{1} {RGB}^{2} {IRGB}^{3} {IRRB}^{4}
Varying Geographic Location	OpenEarthMap	{dev. space, tree}^{1} {rangel., build.}^{2} {agric. land}^{3} {road}^{4} {water}^{5} {barel.}^{6}	{EU}^{1} {AS}^{2} {OC}^{3} {NA}^{4} {AF}^{5} {SA}^{6}
Compound Band-Location Variation	ISPRS	{imperv. surf., build., low veg.}^{1} {tree, car}^{2}	{Vaihingen}^{1} {Potsdam}^{2}

The superscript indicates the learning step. We define three experimental setups based on the causes of concurrent domain shifts in class-incremental learning scenarios.

2) Incremental Learning Setups: To comprehensively assess our framework's performance in class-domain incremental learning, we carefully establish three experimental setups corresponding to different causes of distribution shift. The first experimental setup emulates class-incremental scenarios with concurrent domain shifts induced by varying sensor band configurations. Using the Potsdam dataset, we construct four learning steps by randomly selecting three-channel compositions from the original four spectral bands (IR–R–G–B), while maintaining consistent image splits (17 training, 7 validation, and 14 test images) across all steps. This setup preserves identical spatial coverage while introducing spectral variability between learning steps. For the second setup, we utilize the OpenEarthMap dataset to investigate domain shifts arising from geographical location variations. The incremental learning process is structured into six distinct stages corresponding to continental divisions (Africa, Asia, Europe, North America, South America, and Oceania). For each continental subset, we adopt the 6:1:3 training/validation/test split from [24], using the validation set as the test set since true labels for the original test set are unavailable. The third setup we explore focuses on incremental domain shifts caused by the simultaneous variation of both band configurations and geographical locations. In this setup, we employ the Vaihingen and Potsdam datasets to implement two learning steps. While Vaihingen's official partitioning scheme allocates 11 images for training, 5 for validation, and 17 for testing, we merge the training and validation sets (16 images) due to insufficient training samples, retaining the original 17 test images for evaluation. The Potsdam dataset maintains the same partitioning structure as employed in the first experimental setup. Table I details these three experimental setups for class-domain incremental semantic segmentation in RS.

3) Comparative Methods: Research on class-domain incremental semantic segmentation remains scarce, with LwS being the only one currently available. To ensure a sufficient comparison, we additionally select state-of-the-art methods from class-incremental (LwF [35], MiB [32], NeST [36], and MiSSNet [13]) and domain-incremental (MDIL [7]) semantic segmentation, all with publicly available implementations. Following standard practices in incremental learning studies

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TABLE II
RESULTS (MIOU IN %) ON THE POTSDAM DATASET WITH THE SETUP OF CHANGING SENSOR BAND CONFIGURATION

Method	Seg. Model	Step 1 (IRRG) mIoU_1^1	Step 2 (RGB) mIoU_2^2	Step 2 IRRG mIoU_1^2	mIoU^2	Step 3 (IRGB) mIoU_3^3	Step 3 RGB mIoU_2^3	Step 3 IRRG mIoU_1^3	mIoU^3	Step 4 (IRRB) mIoU_4^4	Step 4 IRGB mIoU_3^4	Step 4 RGB mIoU_2^4	Step 4 IRRG mIoU_1^4	mIoU^4
FT	ERFNet	85.73	43.86	15.96	29.91	39.08	22.09	27.95	29.71	22.59	21.69	20.64	22.53	21.86
MDIL [7]	ERFNet	60.33	39.74	36.95	38.34	34.99	33.01	29.78	32.60	31.98	30.25	28.42	31.95	30.65
LwS [17]	ERFNet	85.10	77.55	67.69	72.62	70.92	74.52	68.77	71.40	70.96	73.88	74.37	71.61	72.71
CDISeg (**ours**)	ERFNet	84.80	77.53	74.93	76.23	71.55	73.77	74.60	73.31	72.57	72.96	73.97	74.77	73.57
LwF [35]	D-R101	89.42	71.25	45.34	58.29	50.01	53.45	43.92	49.12	45.84	51.99	52.76	46.37	49.24
MiB [32]	D-R101	88.81	73.09	73.57	73.33	70.83	70.74	72.41	71.33	74.81	72.53	68.69	74.89	72.73
NeST [36]	D-R101	**89.43**	76.17	65.19	70.68	70.74	70.81	65.99	69.18	72.03	72.76	70.86	69.47	71.28
MiSSNet [13]	D-R101	86.72	73.82	65.49	69.66	70.19	69.91	63.17	67.76	68.20	68.48	66.23	64.84	66.94
LwS [17]	D-R101	84.83	79.09	69.24	74.17	70.97	**76.11**	69.54	72.21	73.12	73.43	**76.39**	72.29	73.81
CDISeg (**ours**)	D-R101	85.60	**79.22**	**76.65**	**77.94**	**76.08**	75.73	**76.37**	**76.06**	**76.95**	**76.86**	74.80	**77.07**	**76.42**
Joint	ERFNet	85.68	80.11	80.48	80.30	78.33	78.03	78.48	78.28	78.92	78.97	78.69	79.01	78.90

Parentheses in the second row denote the training domain per step. *Seg. Model* is short for segmentation model, which includes ERFNet [37] and Deeplab-v3 [38] with ResNet-101 [39] backbone (denoted as D-R101). mIoU_m^t denotes segmentation accuracy on domain m for model t (all classes up to t). mIoU^t is the average over all domains seen up to t. Best and second-best results are in **bold** and underline, respectively.

[17], [32], we include two fundamental baselines: simple fine-tuning (FT) applied sequentially to each learning step and joint optimization (Joint) trained on all classes and domains simultaneously, with the latter serving as the performance upper-bound. We adopt the mean intersection-over-union (mIoU) as our primary performance metric. Specifically, mIoU_m^t denotes the segmentation accuracy of the tth model evaluated on domain m, computed over all classes accumulated from step 1 to t. Meanwhile, mIoU^t represents the model's mean performance averaged over all domains encountered up to step t.

4) Implementation Details: Following common practice in incremental semantic segmentation research, we employ two segmentation models: ERFNet [37] and Deeplab-v3 [38] with a ResNet-101 [39] backbone. The style encoder and reconstruction module adopt the structure from [29], with modified depth and width to match the segmentation encoder's outputs. Our fusion module is composed of a 3×3 convolutional layer, followed by AdaIN and ReLU activation. Its lightweight design minimizes, though does not eliminate, the increase in inference overhead of the segmentation model.

In all experiments, the segmentation models are initialized with ImageNet [40] pretrained weights, while other modules are randomly initialized at the first learning step. For incremental steps ($t > 1$), the framework is initialized using weights from the previous step. When training with ERFNet, we adopt the Adam optimizer with a learning rate of $1.5e^{-3}$ and a batch size of 48. For Deeplab-v3, we use the SGD optimizer with a learning rate of $1e^{-3}$ and a batch size of 2. Weight decay is set to $1e^{-4}$, and we apply a polynomial learning rate decay with a power of 0.9. For all datasets, original images are cropped to 512×512. The models are trained for 50 epochs on the Potsdam dataset, 100 epochs on the Vaihingen dataset, and 50 epochs on the OpenEarthMap dataset. Baseline results are reproduced using the authors' GitHub code, with epochs adjusted to align with that of our model training, while keeping other hyperparameters at their default settings.

B. Comparative Experiments

1) Comparison on the Potsdam Dataset: Results are reported in Table II. When FT the segmentation model by solely optimizing the segmentation task loss for learning new classes on the new domain, the model suffers from catastrophic forgetting of old-class knowledge from past domains. Moreover, the distribution shift between old- and new-domain data prevents the identification of new-class objects in old-domain images. Consequently, FT exhibits poor performance, achieving just 21.86% mIoU^4 after four incremental learning steps. MDIL isolates partial encoder parameters and employs dedicated decoders for new domain learning, mitigating the forgetting issue by freezing domain-specific parameters. During inference, network parameter activation is determined by the input domain. For class-domain incremental semantic segmentation, our implementation activates all decoders concurrently and merges their predictions via the rule: if all predictions unanimously yield background, the result is background; otherwise, the foreground class with the highest probability is selected. Compared to FT, MDIL enhances model capability in preserving old-domain knowledge, ultimately achieving an 8.79% improvement in mIoU^4. However, MDIL's dynamic architecture strategy struggles to effectively learn new classes from the new domain, particularly at the initial learning step, where its performance lags behind FT by over 25%.

Knowledge distillation-based class-incremental learning methods (LwF, MiB, NeST, and MiSSNet) demonstrate superior performance by effectively balancing new class acquisition with old class preservation. Nevertheless, these methods are fundamentally limited by their inability to perform cross-domain knowledge transfer (i.e., extending newly learned classes to previous domains while adapting old classes to the new domain). Consequently, their overall performance remains suboptimal, with MiB achieving the highest score of 72.73%. By synthesizing old-domain images for cross-domain knowledge accumulation, LwS surpasses both class- and domain-incremental learning methods, obtaining 72.71%

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TABLE III

RESULTS (MIOU IN %) ON THE OPENEARTHMAP DATASET WITH THE SETUP OF VARYING GEOGRAPHIC LOCATION

Method			FT	MDIL [7]	LwS [17]	CDISeg (**ours**)	LwF [35]	MiB [32]	NeST [36]	MiSSNet [13]	LwS [17]	CDISeg (**ours**)	Joint
Seg. Model			ERFNet				D-R101						ERFNet
Step 1	(EU)	$mIoU^1_1$	62.21	34.07	59.60	57.65	**66.32**	65.74	64.84	56.91	54.91	56.64	61.58
Step 2	(AS)	$mIoU^2_2$	31.32	33.22	54.45	**54.52**	49.83	53.13	52.57	48.44	48.81	52.84	63.31
	EU	$mIoU^2_1$	25.35	33.50	52.03	52.29	53.85	**57.32**	56.50	52.23	45.78	49.98	63.71
	$mIoU^2$		28.34	33.36	53.24	53.41	51.84	**55.22**	54.54	50.34	47.30	51.41	63.51
Step 3	(OC)	$mIoU^3_3$	14.24	24.42	46.58	48.66	42.62	50.87	**52.53**	44.57	34.75	47.67	67.24
	AS	$mIoU^3_2$	8.35	24.49	44.52	44.80	37.48	**45.70**	45.54	41.50	31.46	44.35	67.17
	EU	$mIoU^3_1$	8.36	22.72	40.54	44.79	40.27	46.71	**51.37**	46.41	28.77	43.96	66.65
	$mIoU^3$		10.32	23.88	43.88	46.08	40.12	47.76	**49.81**	44.16	31.66	45.33	67.02
Step 4	(NA)	$mIoU^4_4$	12.72	23.85	39.99	**41.92**	33.91	37.79	39.46	37.46	33.68	40.22	66.39
	OC	$mIoU^4_3$	11.73	22.70	45.88	48.11	41.34	45.01	43.88	43.37	39.05	**48.76**	65.67
	AS	$mIoU^4_2$	9.79	22.67	41.91	42.71	35.70	40.14	38.89	39.64	33.85	**43.48**	65.64
	EU	$mIoU^4_1$	11.10	22.48	40.05	43.85	38.76	43.48	43.23	**44.99**	31.75	44.77	65.30
	$mIoU^4$		11.34	22.92	41.96	44.15	37.42	41.60	41.37	41.37	34.58	**44.31**	65.75
Step 5	(AF)	$mIoU^5_5$	7.54	20.92	34.08	39.11	28.96	33.64	32.87	31.82	30.80	**39.70**	62.35
	NA	$mIoU^5_4$	2.97	18.36	31.77	**37.20**	29.59	30.67	34.36	35.01	28.05	35.29	66.70
	OC	$mIoU^5_3$	6.36	21.68	37.50	37.51	32.54	39.95	**47.23**	41.59	31.37	44.54	68.22
	AS	$mIoU^5_2$	5.31	19.46	35.30	39.44	30.83	35.80	37.84	37.86	29.97	**42.16**	66.29
	EU	$mIoU^5_1$	3.83	19.43	29.22	34.74	28.90	39.89	38.43	**40.22**	25.87	39.79	68.20
	$mIoU^5$		5.20	19.97	33.57	37.60	30.17	35.99	38.15	37.30	29.21	**40.30**	66.35
Step 6	(SA)	$mIoU^6_6$	3.63	14.91	28.67	31.11	23.85	29.68	31.61	34.65	22.53	**34.77**	59.24
	AF	$mIoU^6_5$	2.18	16.11	23.32	30.13	19.47	25.45	27.32	27.38	20.71	**31.95**	57.14
	NA	$mIoU^6_4$	1.92	13.74	22.75	**31.16**	23.13	28.67	29.25	29.45	15.80	29.33	63.34
	OC	$mIoU^6_3$	0.75	16.14	28.83	29.75	22.85	30.19	**40.07**	34.54	23.11	33.15	59.70
	AS	$mIoU^6_2$	2.31	15.05	23.87	32.51	21.47	29.48	32.16	31.31	18.24	**32.82**	60.42
	EU	$mIoU^6_1$	2.05	15.09	21.55	27.87	20.39	30.75	33.68	**34.01**	16.06	30.66	62.87
	$mIoU^6$		2.14	15.17	24.83	30.42	21.86	29.04	**32.35**	31.89	19.41	32.11	60.45

Parentheses in the second column denote the training domain per step. *Seg. Model* is short for segmentation model, which includes ERFNet [37] and Deeplab-v3 [38] with ResNet-101 [39] backbone (denoted as D-R101). $mIoU^t_m$ denotes segmentation accuracy on domain m for model t (all classes up to t). $mIoU^t$ is the average over all domains seen up to t. Best and second-best results are in **bold** and underline, respectively.

$mIoU^4$ with ERFNet and 73.81% $mIoU^4$ with Deeplab-v3. While LwS shows strong results, our CDISeg further improves upon LwS by **0.86**% (ERFNet) and **2.61**% (Deeplab-v3). It is particularly noteworthy that our framework, when equipped with the lightweight ERFNet, achieves superior results ($mIoU^2$ and $mIoU^3$) or comparable performance ($mIoU^4$) relative to LwS, which relies on the parameter-heavy Deeplab-v3. These results confirm the superiority of feature synthesis in facilitating cross-domain knowledge accumulation.

2) Comparison on the OpenEarthMap Dataset: This setup is challenging due to its extended incremental learning process up to 6 steps. Quantitative results for all competitors and CDISeg are provided in Table III. As anticipated, FT performs extremely poorly, reaching a $mIoU^6$ below 3%, while MDIL outperforms FT by over 13%. Among class-incremental methods, MiSSNet employs a continuously updated feature memory that stores class-specific features from every training image at each learning step, to replay old-domain class knowledge, whereas other methods (LwF, MiB, and NeST) do not store any such data. Even our CDISeg only preserves a single average style feature across the entire training set. To ensure a fair comparison, we limit MiSSNet's old-class feature storage to the averaged class features across all training images per incremental step.

The OpenEarthMap dataset contains RS images from 44 countries, resulting in highly diverse data distributions even within a single learning step (i.e., intracontinent). This intrinsic diversity, in turn, promotes generalization across steps and, crucially, minimizes the domain gap between them. With a small domain gap, the primary challenge in incremental learning is largely reduced to overcoming catastrophic forgetting of old classes. Consequently, class-domain incremental segmentation on OpenEarthMap tends to resemble class-incremental segmentation. Given this context, methods designed for class-incremental learning become highly relevant. Notably, NeST and MiSSNet deliver highly competitive results, achieving $mIoU^6$ of 32.35% and 31.89%, respectively. This strong performance stems from their shared emphasis on addressing the background shift issue inherent to class-incremental semantic segmentation. NeST initializes new classifiers by transforming old ones to avoid abrupt encoder modifications during new class learning, whereas MiSSNet innovates by maintaining old-class semantic memory for local semantic distillation and class-specific regularization.

Among methods tailored for class-domain incremental segmentation, and under the same segmentation model, LwS lags behind class-incremental methods by a notable margin of nearly 13%, whereas our CDISeg attains 32.11% $mIoU^6$, outperforming MiSSNet and reaching performance on par with NeST. This significant performance disparity can be attributed to their respective synthesis designs. LwS likely struggles because it utilizes domain statistics to generate both

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TABLE IV

RESULTS (MIOU IN %) ON THE ISPRS DATASET WITH THE SETUP OF THE COMPOUND BAND AND LOCATION VARIATION

Method	Seg. Model	Vaihingen→Potsdam {imperv. surf., build., low veg.}→{tree, car}				Potsdam→Vaihingen {imperv. surf., build., low veg.}→{tree, car}				Vaihingen→Potsdam {tree, car}→{imperv. surf., build., low veg.}			
		Step 1 (Vaihingen) mIoU_1^1	Step 2 (Potsdam) mIoU_2^2	Step 2 Vaihingen mIoU_1^2	mIoU^2	Step 1 (Potsdam) mIoU_1^1	Step 2 (Vaihingen) mIoU_2^2	Step 2 Potsdam mIoU_1^2	mIoU^2	Step 1 (Vaihingen) mIoU_1^1	Step 2 (Potsdam) mIoU_2^2	Step 2 Vaihingen mIoU_1^2	mIoU^2
FT	ERFNet	75.93	38.72	8.30	23.51	80.17	39.33	18.03	28.68	68.85	48.01	18.35	33.18
MDIL [7]		67.10	28.66	14.20	21.43	69.86	24.71	28.47	26.59	23.44	1.29	8.89	5.09
LwS [17]		76.13	54.12	60.63	57.38	79.75	45.86	47.87	46.87	68.37	53.22	23.49	38.36
CDISeg (**ours**)		74.48	**59.94**	58.59	**59.27**	79.09	47.14	54.12	50.63	61.90	56.56	31.82	44.19
LwF [35]	D-R101	80.00	38.98	29.66	34.32	84.60	32.91	13.40	23.16	**76.71**	59.41	27.89	43.65
MiB [32]		79.93	29.34	54.79	42.07	84.64	44.28	52.98	48.63	75.88	60.53	27.06	43.80
NeST [36]		**80.23**	32.39	55.36	43.88	**84.70**	36.94	49.85	43.40	74.93	**60.75**	29.62	45.19
MiSSNet [13]		77.45	47.39	**65.13**	56.26	81.62	35.00	**58.28**	46.64	64.22	58.51	32.35	45.43
LwS [17]		72.54	56.70	47.53	52.12	79.03	50.95	46.49	48.72	62.91	47.28	19.46	33.37
CDISeg (**ours**)		74.51	56.05	54.82	55.44	79.38	**54.74**	58.26	**56.50**	62.14	57.97	**42.68**	**50.33**
Joint	ERFNet	76.50	78.93	72.34	75.64	80.40	72.34	78.93	75.64	66.10	78.93	72.34	75.64

Parentheses in the fourth row denote the training domain per step. *Seg. Model* is short for segmentation model, which includes ERFNet [37] and Deeplab-v3 [38] with ResNet-101 [39] backbone (denoted as D-R101). mIoU_m^t denotes segmentation accuracy on domain m for model t (all classes up to t). mIoU^t is the average over all domains seen up to t. Best and second-best results are in **bold** and underline, respectively.

old- and new-domain images for model training. These domain statistics capture an average style of the per-step training data. Using this averaged style to synthesize new-domain images diminishes the diversity of the original data distribution, consequently compromising cross-step generalization. Likewise, the old-domain images synthesized from old-domain statistics exhibit constrained diversity, which impairs the model's ability to retain knowledge of old classes. Conversely, our CDISeg benefits from the feature synthesis strategy that randomly alternates between averaged style features and instance-specific style features from the new domain. This hybrid mechanism preserves the natural diversity of the training data and enhances generalization across learning steps. Moreover, our CaSR generates diversified stylized images that are leveraged to reconstruct old-domain features, alleviating the diversity degradation caused by reliance on averaged style characteristics during old class rehearsal.

3) Comparison on the ISPRS Dataset: In this setup, the model, despite its learning consisting of merely two steps, handles substantial domain shift due to compound variations in band configurations and imaging locations. To thoroughly investigate the potential of existing methods and CDISeg in class-domain incremental learning, we vary the learning order of classes and domains based on the configuration specified in Table I. For clarity, we adopt the notation V3 →P2, P3 →V2, and V2 →P3 to represent the different learning sequences on the ISPRS dataset. The left and right sides of the arrow denote the domain used in step 1 and step 2, respectively: Vaihingen (V) and Potsdam (P). The numbers indicate the class sets learned in each step: 3 corresponds to {*impervious surface, building, low vegetation*}, and 2 refers to {*tree, car*}.

The results are presented in Table IV. Both FT and MDIL still yield inferior results, with the latter performing even worse than FT in three experimental setups. All other comparative methods (except LwF) surpass them by a large margin across all setups. Notably, LwS with ERFNet attains the best performance (57.38% mIoU^2) on V3 →P2, LwS with Deeplab-v3 reaches 48.72% on P3 →V2, and MiSSNet achieves 45.43% on V2 →P3. Despite these strong results, CDISeg provides further improvements in all setups. In particular, when compared under the same segmentation model, CDISeg outperforms the strongest competitor, LwS, across all three setups, with the most significant margin approaching **17%** (observed on V2 →P3 using Deeplab-v3). Furthermore, even with a lightweight segmentation model (ERFNet), CDISeg already surpasses MiSSNet on V3 →P2 and P3 →V2, although MiSSNet uses a heavier model (Deeplab-v3). On V2 →P2, CDISeg equipped with Deeplab-v3 exceeds MiSSNet by nearly **5%**. These results demonstrate the effectiveness of CDISeg in handling joint class and domain incremental learning.

It is especially interesting to examine model performance under different learning orders. Regarding domain ordering (comparing V3 →P2 and P3 →V2), all competitors exhibit significant performance fluctuations. For example, LwS with Deeplab-v3 drops from 56.70% to 46.49% on Potsdam, a decline of 10.21%, whereas MiSSNet sharply decreases from 65.13% to 35.00% on Vaihingen, reflecting a gap exceeding 30%. In contrast, CDISeg with Deeplab-v3 maintains consistent performance: improving from 56.05% to 58.26% on Potsdam and remaining stable around 54.74% to 54.82% on Vaihingen. In terms of class ordering (comparing V3 →P2 and V2 →P3), all methods, including CDISeg, show performance fluctuations. However, CDISeg with Deeplab-v3 exhibits the smallest variance: the difference on Vaihingen is only 1.92%, and on Potsdam it is 12.14%. These values are considerably lower than those of LwS (9.42% and 28.07%) and MiSSNet (11.12% and 32.78%). These results confirm that CDISeg offers more robust and reliable incremental learning capability compared to existing alternatives.

4) Visualization: To illustrate class-domain incremental semantic segmentation intuitively, we present some qualitative results on the ISPRS dataset, as depicted in Fig. 6. Different

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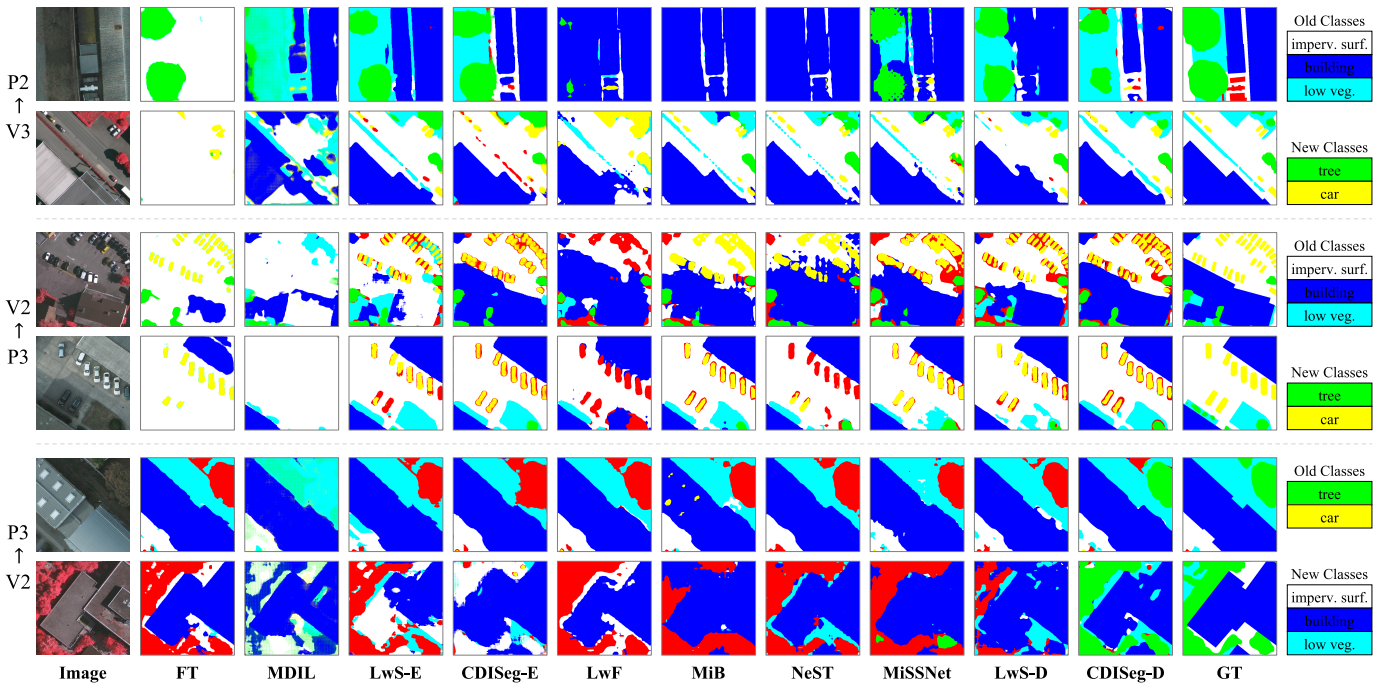


Fig. 6. Qualitative results for different learning orders (V3 →P2, P3 →V2, and V2 →P3) on the ISPRS dataset. Different colors, that is, white, blue, cyan, green, yellow, and red, indicate impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background, respectively. The suffixes (−E and −D) in LwS and CDISeg denote the use of ERFNet and Deeplab-v3 as their segmentation models. GT is short for ground truth.

colors are used to indicate impervious surfaces (white), buildings (blue), low vegetation (cyan), trees (green), cars (yellow), and clutter/background (red). On the V3 →P2 incremental setup, *tree* and *car* are new classes learned from the Potsdam subset. The difference in imaging band configurations between the two datasets leads to noticeable visual disparities for the *tree* class in Vaihingen versus Potsdam images. As a result, methods such as FT, MDIL, and LwF fail to recognize trees in the Vaihingen domain. On the other hand, *building* and *low vegetation* are previously learned classes from the Vaihingen subset. Among all methods, only our CDISeg (both -E and -D variants) achieves relatively accurate segmentation of buildings and low vegetation in the Potsdam image. This result confirms that leveraging synthetic features enables the model to not only adapt old classes to the new domain but also effectively extend new classes to past domains.

The P3 →V2 setup represents a reversal of the learning domain compared to V3 →P2. As illustrated in the third row of Fig. 6, only FT, LwS-D, and CDISeg-D are effective on new classes within the new domain (Vaihingen). However, the fourth row reveals that among these, only CDISeg-D successfully identifies old classes in the previous domain (Potsdam), particularly the *low vegetation* class. This highlights CDISeg's capacity to balance stability with plasticity, a critical capability for incremental learning. On the final V2 →P3 setup, limited training data from Vaihingen hinder the learning of *tree* and *car* classes within the domain, while also compromising knowledge retention and transfer to subsequent domains. Consequently, nearly all incremental methods, except CDISeg-D, succumb to catastrophic forgetting, failing to recognize trees in either Vaihingen or Potsdam. Regarding new classes, all methods except MiB, MDIL, and LwS-E perform satisfactorily on Potsdam. However, only NeST and CDISeg-D manage to yield buildings on Vaihingen that approach the ground truth.

C. Ablation Studies

We conduct ablation experiments under the most challenging setup on the ISPRS dataset, employing ERFNet as the segmentation model unless noted otherwise, to verify the impact of our framework's design details.

1) Effectiveness of Feature Synthesis and Style Randomization: Table V reports contributions of CDISeg's design details. We start from the baseline MiB [32] that employs cross-entropy and knowledge distillation losses accounting for the background shift. MiB assigns a higher weight to the knowledge distillation loss by default, encouraging the segmentation model to retain old-class knowledge during incremental learning. However, this implementation partially inhibits learning of new classes, leading to a performance of 61.29% on Vaihingen ($\mathcal{C}^1$) but only 21.42% on Potsdam ($\mathcal{C}^2$). Furthermore, in the absence of additional constraints, the distribution shift between the Potsdam and Vaihingen datasets hinders the model's ability to recognize old-class objects in new-domain images (34.47%) and new-class objects in old-domain images (22.51%).

We integrate into the baseline our feature synthesis, along with two losses designed for cross-domain class knowledge transfer (+*FeatSyn*). This substantially boosts the model's performance at step 2, yielding gains of **24.94%** mIoU_2^2 on Potsdam and **6.45%** mIoU_1^2 on Vaihingen. Specifically, the old-domain features, synthesized from new-domain content and old-domain style, not only facilitate the preservation of old

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TABLE V

EVALUATION OF FEATURE SYNTHESIS AND CASR ON THE ISPRS DATASET WITH THE SETUP OF THE COMPOUND BAND AND LOCATION VARIATION

Method		MiB [32]	+FeatSyn	+CaSR
Step 1 (Vaihingen)	mIoU_1^1	**75.73**	74.41 (-1.32)	74.48 (+0.07)
Step 2 (Potsdam)	$\mathcal{C}^1$	34.47	41.19 (+6.72)	**52.59** (+11.40)
	$\mathcal{C}^2$	21.42	**73.68** (+52.26)	70.96 (-2.72)
	mIoU_2^2	29.25	54.19 (+24.94)	**59.94** (+5.75)
Step 2 Vaihingen	$\mathcal{C}^1$	61.29	62.91 (+1.62)	**67.02** (+4.11)
	$\mathcal{C}^2$	22.51	36.20 (+13.69)	**45.94** (+9.74)
	mIoU_1^2	45.78	52.23 (+6.45)	**58.59** (+6.36)
mIoU^2		37.52	53.21 (+15.69)	**59.27** (+6.06)

FeatSyn disentangles image content and style features via orthogonal constraints, then synthesizes old-domain features for cross-domain knowledge accumulation. *CaSR* improves disentanglement by style randomization. $\mathcal{C}^1$ and $\mathcal{C}^2$ denote the mIoU computed on the semantic categories learned in the first and second steps, respectively. mIoU_m^t denotes segmentation accuracy on domain m for model t (all classes up to t). mIoU^t is the average over all domains seen up to t. Best results are in **bold**, and improvements are shown in (red).

classes (61.29% versus 62.91%) but also support the extension of new classes to past domains (**22.51% versus 36.20%**). Meanwhile, by leveraging synthesized features, the old model can more accurately recognize old-class objects in new-domain images, thereby effectively guiding the new model to adapt old classes to the new domain. This results in a notable improvement of **6.72%** on Potsdam ($\mathcal{C}^1$). Additionally, the introduction of cross-domain knowledge transfer losses ($\mathcal{L}_{kd}^n$ and $\mathcal{L}_{ce}^o$) with the same weight as the original knowledge distillation loss ($\mathcal{L}_{kd}^o$) mitigates the new model's bias toward old classes from old domains, achieving an improvement of over **52%** for new classes in the new domain. These results demonstrate that feature synthesis-based cross-domain knowledge accumulation effectively enables the model to perform joint class and domain incremental learning.

Finally, we add CaSR to enhance feature disentanglement (+CaSR). This design leads to remarkable performance gains, including improvements of **4.11%** on old classes and **9.74%** on new classes in the Vaihingen dataset, as well as **11.40%** on old classes in the Potsdam dataset. Unexpectedly, however, the model's performance on new classes in the new domain experienced a slight decline (−2.72%). This drop occurs because the feature disentanglement constraint inevitably impairs the content encoder's ability to learn discriminative features for semantic recognition. This effect is especially evident in step 1, where both +FeatSyn and +CaSR slightly underperformed the baseline. Furthermore, to minimize computational overhead during inference, we use an extremely lightweight fusion module, which also contributes to information loss and hinders the learning of new classes in the new domain. Nevertheless, enhanced feature disentanglement promotes more meaningful feature separation and yields higher-quality old-domain features, leading to a substantial overall performance improvement, with mIoU^2 increasing from 53.21% to 59.27%. This confirms the importance of improving feature disentanglement.

TABLE VI

CLASS-WISE PIXEL PROPORTION (IN %) ACROSS CONTINENTAL SUBSETS OF THE OPENEARTHMAP TRAINING DATASET

Continent	EU	AS	OC	NA	AF	SA
# Train Imgs	2,936	2,128	672	2,428	2,040	1,670
Barel.	0.33	0.59	0.39	1.34	1.85	5.68
Rangel.	21.52	17.72	24.56	29.14	24.35	14.90
Dev. Space	12.98	15.91	13.27	12.04	19.25	27.35
Road	6.15	5.75	7.06	9.41	5.23	7.62
Tree	21.67	19.44	14.98	24.85	19.56	10.73
Water	3.13	4.17	2.98	2.95	4.40	1.36
Agric. Land	19.95	17.47	22.75	6.92	6.21	11.55
Build.	13.59	18.16	13.99	12.96	18.47	19.90

EU, *AS*, *OC*, *NA*, *AF* and *SA* denote Europe, Asia, Oceania, North America, Africa, and South America, respectively.

TABLE VII

DIFFERENT LEARNING ORDERS ON THE OPENEARTHMAP DATASET

ID	Learning Order
①	EU→AS→OC→NA→AF→**SA**
②	AS→OC→EU→NA→AF→**SA**
③	AS→EU→OC →NA→AF→**SA**
④	{dev. space, tree} → {rangel., build.} → {agric. land} → {road} → {water} →{**barel.**}
⑤	{dev. space, build.} → {tree} → {rangel., agric. land} → {road} → {water} → {**barel.**}
⑥	{tree, agric. land} → {dev. space, build.} → {rangel.} → {road} → {water} → {**barel.**}

Bold text denotes the semantic class or domain acquired at the final incremental learning step. Experiments ① to ③ follow the category learning order specified in Table I, whereas experiments ④ to ⑥ employ the domain learning order from the same table.

2) Evaluation With Different Learning Orders: As depicted in Table IV, CDISeg demonstrates superior performance compared to existing methods across different class-incremental and domain-incremental orders. To thoroughly investigate the impact of learning orders on our framework, we conduct experiments using the OpenEarthMap dataset, as it offers the most learning steps. Table VI presents class-wise proportion across continental subsets of the OpenEarthMap training dataset, highlighting a pronounced class imbalance. In practical applications, data newly acquired for novel class recognition inherently contains a substantial number of samples belonging to these target categories. We therefore jointly consider both training data volume and class proportions, associating specific continents with classes where they are most prevalent (e.g., bareland in South America and road in North America). Building on this pairing strategy, we vary the learning order across the remaining classes and domains, designing distinct sequences as detailed in Table VII.

The performance of CDISeg under different learning orders is reported in Table VIII. The results clearly indicate a strong dependence of model performance on the sequence of learning, with a significant gap of 4.05% in mIoU^6 between the best result (Order 2, 31.93%) and the worst (Order 3, 27.88%). Among the different learning orders, the variations mainly emerge in the first three steps. A comparison of Orders 1, 2,

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TABLE VIII

RESULTS (MIOU IN %) OF DIFFERENT LEARNING ORDERS ON THE OPENEARTHMAP DATASET WITH THE SETUP OF VARYING GEOGRAPHIC LOCATION

ID		①	②	③	④	⑤	⑥
Step 1	mIoU^1_1	57.65	58.74	59.94	57.65	62.43	76.57
Step 2	mIoU^2_2	54.52	53.60	53.21	54.52	57.44	58.57
	mIoU^2_1	52.29	49.22	52.33	52.29	58.95	56.65
	mIoU^2	53.41	51.41	52.77	53.41	58.20	57.61
Step 3	mIoU^3_3	48.66	47.35	46.62	48.66	61.48	59.12
	mIoU^3_2	44.80	54.44	44.50	44.80	44.61	49.40
	mIoU^3_1	44.79	49.18	41.34	44.79	51.04	51.37
	mIoU^3	46.08	50.32	44.15	46.08	52.38	53.30
Step 4	mIoU^4_4	41.92	47.16	41.80	41.92	48.20	51.31
	mIoU^4_3	48.11	45.66	47.26	48.11	58.95	57.26
	mIoU^4_2	42.71	52.32	43.82	42.71	41.53	45.45
	mIoU^4_1	43.85	45.90	40.45	43.85	47.46	47.07
	mIoU^4	44.15	47.76	43.33	44.15	49.04	50.27
Step 5	mIoU^5_5	39.11	40.17	36.23	39.11	38.70	40.31
	mIoU^5_4	37.20	41.61	34.92	37.20	39.48	42.72
	mIoU^5_3	37.51	36.64	37.00	37.51	48.94	45.65
	mIoU^5_2	39.44	46.14	34.53	39.44	38.01	42.45
	mIoU^5_1	34.74	43.66	37.82	34.74	38.03	36.24
	mIoU^5	37.60	41.64	36.10	37.60	40.63	41.47
Step 6	mIoU^6_6	31.11	34.10	28.24	31.11	31.73	32.81
	mIoU^6_5	30.13	30.22	27.04	30.13	30.11	30.54
	mIoU^6_4	31.16	32.94	28.65	31.16	31.99	32.63
	mIoU^6_3	29.75	27.40	28.47	29.75	36.25	33.82
	mIoU^6_2	32.51	33.51	25.92	32.51	30.73	33.36
	mIoU^6_1	27.87	33.43	28.98	27.87	29.24	26.98
	mIoU^6	30.42	31.93	27.88	30.42	31.68	31.69

The learning orders referred to by the ID are provided in Table VII. mIoU^t_m denotes segmentation accuracy on domain m for model t (all classes up to t). mIoU^t is the average over all domains seen up to t.

and 3, which follow the same class learning sequence but differ in domain order, reveals that performance discrepancy likely arises because certain categories in specific domains are more challenging to learn. Deficient acquisition of class Knowledge impedes its effective transfer in subsequent incremental steps. For example, the OC domain contains fewer samples than the EU domain (672 versus 2936). When the *agricultural land* is learned in the OC domain (as in Orders 1 and 3), it yields a lower mIoU^3 (46.08% and 44.15%, respectively) compared to when the same category is learned in the EU domain (as in Order 2), where the mIoU^3 reaches 50.32%.

A similar trend is observable in the experiments on class ordering, which maintain an identical domain learning sequence but vary in class order. Compared to Order 4, Order 5 achieves a higher mIoU^3 (52.38% versus 46.08%) by incorporating more categories from the OC domain, thus enriching the diversity of training data. Meanwhile, Order 6 markedly improves mIoU^3 (reaching 53.30%) by prioritizing the learning of the *agricultural land*, which has higher pixel prevalence in the EU domain, over the *developed space*. This adjustment allows the model to acquire more representative class knowledge.

TABLE IX

EVALUATION OF λ_{ORTH} AND λ_{REC} ON THE ISPRS DATASET WITH THE SETUP OF THE COMPOUND BAND AND LOCATION VARIATION

Weight Values		Step 1 (Vaihingen) mIoU^1_1	Step 2 (Potsdam) mIoU^2_2	Step 2 Vaihingen mIoU^2_1	mIoU^2
$\lambda_{\text{orth}} = 20$		72.69	59.34	57.82	58.58
$\lambda_{\text{orth}} = 15$		74.04	59.96	61.61	60.79
$\lambda_{\text{orth}} = 10$	$\lambda_{\text{rec}} = 0.01$	74.48	59.94	58.59	59.27
$\lambda_{\text{orth}} = 5$		76.39	58.73	60.38	59.56
$\lambda_{\text{orth}} = 1$		76.01	59.21	62.00	60.61
$\lambda_{\text{orth}} = 10$	$\lambda_{\text{rec}} = 0.1$	73.69	57.56	61.07	59.32
	$\lambda_{\text{rec}} = 0.05$	74.53	57.52	60.11	58.82
	$\lambda_{\text{rec}} = 0.01$	74.48	59.94	58.59	59.27
	$\lambda_{\text{rec}} = 0.005$	74.70	59.70	57.68	58.69
	$\lambda_{\text{rec}} = 0.001$	74.75	59.00	58.74	58.87

λ_{orth} and λ_{rec} are weighting factors for content-style orthogonality and stylized-original content consistency, respectively. For all experiments in this article, λ_{orth} equals 10 and λ_{rec} is 0.01. Parentheses in the second row denote the training domain per step. mIoU^t_m denotes segmentation accuracy on domain m for model t (all classes up to t). mIoU^t is the average over all domains seen up to t.

3) Evaluation With Different Segmentation Models: In Tables II–IV, we evaluate CDISeg and LwS using two architectures: the lightweight ERFNet and the more parameter-heavy Deeplab-v3, to assess whether the performance of class-domain incremental learning methods scales with greater model capacity. LwS fails to consistently benefit from advancements in segmentation model capabilities. While its performance improves with Deeplab-v3 on the Potsdam and ISPRS (P3 →V2) experimental setups, it significantly deteriorates on the OpenEarthMap and ISPRS (V3 →P2, V2 →P3) setups. For instance, on OpenEarthMap, mIoU^6 drops from 24.83% to 19.41%. In contrast, CDISeg demonstrates consistent performance improvements across most experimental setups with model upgrades, despite a lack of gain in the ISPRS (V3 →P2) setup. Furthermore, under multiple setups, CDISeg using ERFNet outperforms LwS, regardless of whether LwS employs Deeplab-v3 or ERFNet. These results demonstrate that CDISeg offers a more generalizable framework for class-domain incremental segmentation of RS images.

4) Parameter Sensitivity: Our CDISeg framework consists of two core designs: cross-domain knowledge accumulation via old-domain feature synthesis and CaSR for improved feature disentanglement. We, therefore, evaluate the impact of weighting factors for content-style orthogonality and stylized-original content consistency, *that is*, λ_{orth} and λ_{rec}. In all comparative experiments, we fix $\lambda_{\text{orth}} = 10$ and $\lambda_{\text{rec}} = 0.01$, with their values determined based on the magnitudes of the associated loss terms. This choice places the cross-domain knowledge accumulation loss and the feature disentanglement/synthesis losses on comparable numerical scales, aligning with our view that both components are equally important throughout incremental learning. As shown in Table IX, we additionally report results under various alternative hyperparameter settings, where CDISeg consistently outperforms state-of-the-art methods across all configurations. The most substantial performance gain (2.53%) is achieved

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TABLE X

EVALUATION OF COMPUTATIONAL COST ACROSS DIFFERENT METHODS ON THE ISPRS DATASET WITH THE SETUP OF THE COMPOUND BAND AND LOCATION VARIATION

Method	#Parameters (M)	GFLOPs	Inference Time (ms/image)
LwF [35]	58.04	67.73	80.15
MiB [32]	58.04	67.73	80.56
NeST [36]	58.04	67.73	42.37
MiSSNet [13]	58.04	67.73	40.77
LwS [17]	58.63	242.13	52.67
CDISeg (**ours**)	104.77	396.77	69.14

All methods adopt Deeplab-v3 as the segmentation model and are tested under identical settings: an image size of 512×512, a batch size of 1, and an NVIDIA GeForce RTX 3090 GPU.

TABLE XI

GENERALIZATION PERFORMANCE AS MIOU COMPUTED ON THE POTSDAM TEST SET UNDER THE IR–G–B BAND CONFIGURATION

Method	V3→P2			P3→V2			V2→P3		
	$\mathcal{C}^1$	$\mathcal{C}^2$	mIoU	$\mathcal{C}^1$	$\mathcal{C}^2$	mIoU	$\mathcal{C}^1$	$\mathcal{C}^2$	mIoU
LwS [17]	45.24	74.78	57.05	56.45	21.14	42.33	13.17	58.88	40.60
CDISeg (**ours**)	51.85	67.14	57.97	59.70	44.56	53.65	28.09	55.22	44.37

The model evaluated for generalization is based on the ERFNet architecture and has completed two incremental learning steps on the ISPRS dataset. V3→P2, P3→V2, and V2→P3 represent different learning sequences. $\mathcal{C}^1$ and $\mathcal{C}^2$ denote the mIoU computed on the semantic categories learned in the first and second steps, respectively.

when $\lambda_{\text{orth}} = 15$ with $\lambda_{\text{rec}} = 0.01$. CDISeg exhibits strong robustness to moderate changes in these hyperparameters, as evidenced by a low standard deviation of 0.81% in mIoU^2 across nine configurations. Furthermore, we note that excessively high values of either λ_{orth} or λ_{rec} can degrade CDISeg's effectiveness, as they cause the model to over-optimize feature disentanglement, ultimately at the expense of its primary category-discriminative learning objective. While reducing λ_{orth} only marginally affects performance, lowering λ_{rec} substantially compromises the quality of synthesized old-domain features, causing a significant drop in old-domain performance.

5) Analysis of Computational Cost: We compare model size (number of parameters), computational complexity (in GFLOPs), and inference time (in milliseconds per image) between CDISeg and strong competitors to evaluate its computational cost. All models are tested under identical settings: an image size of 512×512, a batch size of 1, and an NVIDIA GeForce RTX 3090 GPU. As shown in Table X, CDISeg exhibits larger model size and higher computational complexity than NeST, MiSSNet, and LwS, due to the incorporation of the fusion module, even though we intentionally adopt a lightweight design composed of a single convolutional layer followed by AdaIN and ReLU activation. The inference time also increases; however, the per-image latency of 69.14 ms remains practical for real-world applications. Importantly, this additional inference cost translates into substantial performance gains.

6) Analysis of Generalization Performance: Finally, we further assess the generalization capability of the segmentation model after completing two incremental learning steps on the ISPRS dataset, using the Potsdam test set under the IR-G-B band configuration. It is important to note that we cannot evaluate generalization on other datasets because their predefined semantic classes differ from those of the ISPRS dataset, making quantitative assessment infeasible. Since the Potsdam dataset used during incremental learning includes only the R–G–B bands, the IR–G–B configuration can be reasonably regarded as an out-of-distribution variant for testing the model's generalization ability. Results are reported in Table XI. Across all incremental learning sequences, models trained with CDISeg consistently demonstrate stronger generalization than those trained with LwS. The largest gap appears on the P3 →V2 sequence, where the CDISeg-trained model reaches 53.65% mIoU, whereas the LwS-trained model achieves only 42.33%. These results indicate that CDISeg contributes to improved generalization performance. This benefit arises from its feature synthesis: CDISeg repurposes the segmentation encoder to focus on capturing high-level semantics while disregarding low-level visual differences introduced by varying band configurations. Meanwhile, the stored domain-specific style is projected onto the test image's content features, thereby aligning the test features with the feature distribution of the training data. Together, these designs enable the model to achieve superior performance on out-of-distribution test images.

V. CONCLUSION

In this article, we focus on class-domain incremental segmentation for RS images, and propose CDISeg, a novel framework designed to facilitate cross-domain knowledge accumulation through feature synthesis. CDISeg employs orthogonality constraints to disentangle image style and content, thereby preserving domain-specific style features at each incremental learning step. This allows feature synthesis of previous domains by projecting old-domain styles onto the content of the new domain. To improve the quality of synthesized features, CaSR is incorporated. The framework leverages four tailored losses to guide model learning, promoting effective accumulation of class knowledge across both new and past domains. Extensive experiments on the ISPRS and OpenEarthMap datasets demonstrate the superior performance of CDISeg over recent state-of-the-art methods such as LwS and MiSSNet in various incremental setups. Furthermore, ablation studies are conducted to verify the effectiveness of CDISeg's design details.

Limitations. A comparison of CDISeg's performance on the OpenEarthMap and ISPRS datasets reveals that its improvements over existing methods are less pronounced on OpenEarthMap than on ISPRS. One possible reason is that the training data in each incremental step of OpenEarthMap comes from multiple countries or regions within one continent, leading to considerable intrinsic distribution diversity even within a single domain. Since CDISeg preserves only one style feature per domain, it may fail to adequately capture the full stylistic variety of the current domain. This limitation ultimately affects the quality of synthesized old-domain features. Future work will explore more effective strategies for synthesizing information from previous domains or investigate

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alternative methods to cross-domain knowledge accumulation that reduce reliance on synthetic feature generation.

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